1. Clinical Question & Core Hypothesis

Clinical Question

BRCA1-deficient tumors are expected to be sensitive to PARP inhibition and are often genomically unstable and immune-infiltrated, yet not all BRCA1-deficient, TIL-high tumors achieve durable therapy responses. Why does immune infiltration fail to cooperate with PARP inhibitor efficacy in a subset of these tumors? 2) Tier 1 paths used: (a) BRCA1_loss → homologous_recombination_deficiency → PARP_inhibitor_sensitivity (mechanistic backbone linking BRCA1 status to therapy response); (b) Tumor_mutational_burden/genomic_instability → neoantigen_load → increased TIL_density, but effectiveness depends on antigen presentation (HLA genes). I built on these because they directly explain the mismatch between expected sensitivity and observed variable response. 3) Tier 2 evidence supporting the claim: immune-exhaustion and checkpoint markers (PDCD1, LAG3, TIGIT) and DNA-repair modifiers (BRCA2, RAD51) can modulate both immune efficacy and PARP sensitivity; stromal factors (NABA_CORE_MATRISOME genes like COL1A1) can limit immune–tumor contact. These inform secondary_modifiers. 4) Why therapy_response was chosen as the endpoint: the user's query targeted BRCA1→Therapy, and the most relevant clinical puzzle is variable response to PARP inhibitors in BRCA1-deficient breast cancer; therapy response is therefore the problem-driven endpoint. 5) Deliberately excluded: I excluded multi-gene predictive models and broader endpoints (OS/PFI) to keep the hypothesis mechanistic and specific; Tier 3 exploratory genes (e.g., metabolic genes) were left out of the main claim to preserve focus?

Working Hypothesis

In BRCA1-deficient breast cancers, homologous recombination deficiency (HRD) drives increased genomic instability and neoantigen load, eliciting tumor-infiltrating lymphocytes (TILs). However, the anti-tumor efficacy of PARP inhibitors in this context fully relies on effective antigen presentation via HLA-DQB1. I hypothesize that: (a) BRCA1-deficient tumors with high TIL density and high HLA-DQB1 expression will show the strongest PARP inhibitor–related therapy response; (b) BRCA1-deficient tumors with high TIL density but low HLA-DQB1 expression will exhibit attenuated therapy response due to impaired antigen presentation and suboptimal T-cell–mediated cytotoxicity; (c) tumors with high TIL density and adequate HLA-DQB1 may still be limited by T-cell exhaustion (PDCD1, LAG3) and stromal barriers (COL1A1), which can blunt response magnitude. Consequently, the predicted interaction is: high TIL density plus intact HLA-DQB1 yields the greatest PARP inhibitor response, whereas high TIL density with HLA-DQB1 deficiency yields reduced response, even in BRCA1-deficient tumors. Secondary modifiers (BRCA2, RAD51, PDCD1, LAG3, COL1A1) are expected to modulate the magnitude of this interaction via DNA repair efficiency, T-cell function, and tumor microenvironment constraints. Primary endpoint: therapy_response to PARP inhibitors. This hypothesis is testable through integrated analyses of (i) in silico correlates in BRCA1-deficient tumors (TCGA-BRCA and PARP-inhibitor–treated cohorts) stratified by TIL density and HLA-DQB1 expression; (ii) in vitro co-culture assays using BRCA1-deficient breast cancer cell lines and patient-derived TILs with manipulated HLA-DQB1 levels; and (iii) in vivo humanized or syngeneic models to assess tumor shrinkage and immune metrics under PARP inhibition.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	BRCA1, HLA-DQB1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

BRCA1: BRCA1_loss → homologous_recombination_deficiency → PARP_inhibitor_sensitivity (mechanistic backbone linking BRCA1 status to therapy response); (b) Tumor_mutational_burden/genomic_instability → neoantigen_load → increased TIL_density, but effectiveness depends on antigen presentation (HLA genes).

HLA-DQB1: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PARP1	Supporting modifier from knowledge graph
Secondary	BRCA2	Supporting modifier from knowledge graph
Secondary	PDCD1	Supporting modifier from knowledge graph
Secondary	LAG3	Supporting modifier from knowledge graph
Secondary	COL1A1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: capture T-cell exhaustion; BRCA2/RAD51 influence residual DNA repair capacity; COL1A1 reflects stromal barriers that can limit immune-tumor contact. This framework yields a concise, testable mechanism that links genomic instability, tumor immunity, and PARP-inhibitor response, enabling combinatorial biomarker strategies that rely only on the proposed anchor genes (BRCA1, HLA-DQB

Prediction 2: and the imaging feature (TIL density) with a therapy-response endpoint. The approach is feasible using publicly available datasets (TCGA-BRCA, PARP inhibitor cohorts), standard co-culture and gene manipulation techniques, and established in vivo models to validate the interaction and its modifiers.

Prediction 3: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PARP1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- BRCA2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.